# AI for Chemistry Report — Group 3

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## Abstract

Herein, machine learning models to predict peak emission wavelength ($\lambda_{\mathrm{em,max}}$) and brightness of fluorescent proteins from protein sequence (2D) and three-dimensional structure (3D) are reported. The 2D pipeline combines ESM-2 or T-scale protein embeddings with ChemBERTa chromophore embeddings, processed by multilayer perceptrons. The 3D pipeline uses message-passing neural networks over residue- and atom-level graphs. Both models are trained on proteins from FPBase and evaluated via 5-fold cross-validation. $\lambda_{\mathrm{em,max}}$ is predicted with reasonable accuracy (MAE $\sim$ 23–28 nm), while brightness remains difficult to capture, reflecting its dependence on protein–chromophore interactions.

## 1 Introduction

Fluorescent proteins (FPs) are widely used in cell biology to monitor cellular processes with high spatiotemporal resolution. Engineering FPs with tailored photophysical properties, particularly peak emission wavelength ($\lambda_{\mathrm{em,max}}$) and brightness, is therefore of major interest. However, the complex relationship between amino acid sequence, three-dimensional structure, chromophore environment, and fluorescence output makes rational protein design challenging Laino et al. [2004]. Recent advances in machine learning for protein science, including models such as ProteinMPNN Dauparas et al. [2022] and AlphaFold Jumper et al. [2021], demonstrate the potential of AI-driven approaches for learning complex sequence–structure relationships. In this work, machine learning models combining multilayer perceptrons (MLPs) and message-passing neural networks (MPNNs) Gilmer et al. [2017] to predict FP emission wavelength and brightness from both sequence-based and structure-based protein representations were investigated.

## 2 Task

The goal of this work is to develop both a 2D sequence-based model and a 3D structure-based model for predicting FP emission wavelength and brightness. While fluorescence is primarily governed by the chromophore, its optical properties are strongly modulated by the surrounding protein environment through local and long-range interactions Megley et al. [2009], Laino et al. [2004]. Capturing these effects explicitly remains difficult for machine learning models. It is therefore investigated whether sequence information and/or full three-dimensional structural representations can be used to identify predictive features associated with fluorescent protein photophysics.

## 3 Data

For this project, the FPBase database was used as a curated source of FPs to construct a standardized dataset for subsequent machine learning tasks Lambert [2019]. FPBase was selected because it is open-source and contains experimentally measured photophysical properties in a consistent format, including brightness and $\lambda_{\mathrm{em,max}}$ for each protein. Using the FPBase GraphQL API, protein names, excitation and emission wavelengths, extinction coefficients, quantum yields, brightness values, and FPBase IDs were retrieved, while removing incomplete entries. The resulting data set was then manually processed further to (1) keep only proteins with solved 3D crystal structures and (2) identify

AI4Chemistry course project at EPFL, 2026. https://github.com/schwallergroup/ai4chem_course

the chromophore information available in the Protein Data Bank (PDB) for each protein, resulting in $N = 122$ proteins. For the 3D structure-based model, structures were parsed from PDB files to extract only the relevant structural elements (i.e., $C_\alpha$ atom of each residue and the chromophore/ligand atoms). For each protein, only residues from chain A, and alternate location A were taken if ambiguous. The chromophore region was extracted using predefined residue identifiers and converted into a small PDB block suitable for RDKit processing. Proteins that did not have their $C_\alpha$ atom positions resolved were removed, resulting in a dataset size of N=118 for the 3D-based model.

# 4 Methodology

## 4.1 2D Representation

In the 2D model, only the sequences of each protein are input. The overall workflow is illustrated in Figure 1. T-scale, a descriptor set that associates each amino acid with a unique five-row vector, is tested first. This very general representation was derived by principal component analysis of 67 topological descriptors for 135 amino acids, retaining the five most important features Tian et al. [2007]. Although a purely two-dimensional description does not account for secondary or tertiary protein structures, it has been demonstrated to be a powerful representation tool. For instance, Saito et al. [2018] successfully trained a machine-learning model of the pre-NN generation on matrices obtained by applying T-scale to mutated forms of GFP. This model was able to predict meaningful changes of color upon single-amino acid mutations in GFP.

As a second, more complex representation, ESM-2, a protein language model developed by Lin et al. [2023], was used. This open-source model was trained on a set of 250 million protein sequences in a self-supervised fashion. ESM-2 learns to encode evolutionary and structural information implicitly, without needing exact structure data as input. In this work, a 650-million-parameter variant (ESM2_t33_650M_UR50D) was chosen, which produces a per-residue embedding matrix. To obtain a fixed-length representation, mean-pooling is applied throughout the residue dimension to obtain a single 1280-dimensional vector per protein. Unlike T-scale encoding, ESM-2 embeddings are context-dependent; each residue's representation is influenced by all others in the sequence, potentially capturing long-range interactions relevant to photophysical properties.

Chromophore information was incorporated into model inputs, given its central role in governing fluorescent protein photophysics. Each chromophore was represented by its SMILES string, encoded using the transformer-based model ChemBERTa-3 from Singh et al. [2026], pre-trained on the ZINC molecular database. By mean pooling, the model produces contextual token-level embeddings with a hidden dimension of 768 per SMILES, which corresponds to the size of hidden layers in chemBERTa inherited from the RoBERTa model from Devlin et al. [2019]. To obtain a fixed-length representation independent of SMILES length, mean pooling was applied across the sequence dimension, yielding a single 768-dimensional embedding vector per chromophore. This pooling strategy ensures compatibility with variable-length molecular strings while preserving the global chemical context encoded by the transformer.

Protein embeddings were concatenated with ChemBERTa SMILES embeddings and two experimental scalar features, yielding input vectors of 2,050-d and 1,026-d for the ESM-2 and T-scales

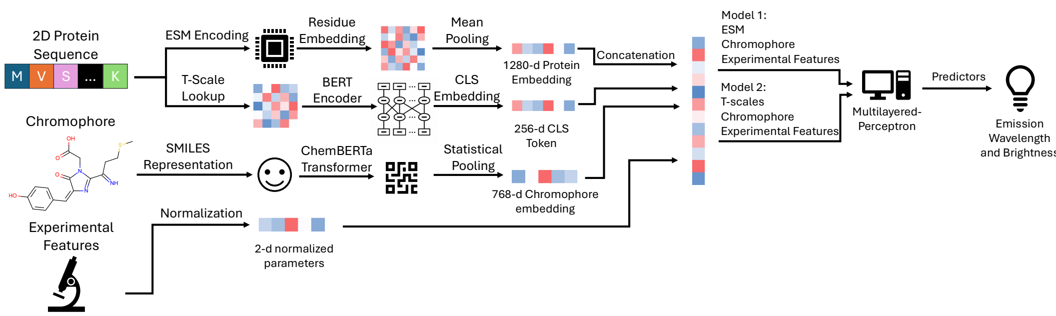


Figure 1: Overview of the machine learning pipeline for the 2D-based model.

pipelines, respectively. Feature vectors and joint target variables ($\lambda_{\mathrm{em,max}}$, brightness) were standardized using statistics computed exclusively from the training fold to prevent data leakage.

Two MLPs, one per encoding scheme, were trained using 5-fold cross-validation over 10 random seeds. At each fold, a grid search across 15 architectures (1–3 hidden layers, 16–256 neurons) selected the configuration minimizing validation RMSE. Final metrics were computed on aggregated out-of-fold predictions.

### 4.2 3D representations

As an alternative representation of proteins, a unified graph-based neural architecture (FPNet) is proposed to predict fluorescent protein photophysical properties from structural (3D coordinates) and sequence-derived information. The workflow is illustrated in Figure 2. Each protein is represented using two complementary graph views: a residue-level protein contact graph (ProteinMPNN) and an atom-level chromophore graph (ChromMPNN), inspired by Schwaller et al. [2026].

The protein is modeled as a residue-level graph where nodes correspond to amino acids positioned at their $C_\alpha$ atoms, and edges connect residues within an 8 Å $C_\alpha$–$C_\alpha$ distance cutoff, forming a bidirectional graph without self-loops. Nodes are one-hot encoded amino acid types, and edge attributes encode Euclidean distances. In parallel, the chromophore is represented as an atom-level graph derived from its Chemical Component Dictionary Westbrook et al. [2015] entry, where nodes correspond to heavy atoms and edges represent covalent bonds with bidirectional connectivity. Atom features consist of 9-dimensional RDKit–OGB Hu et al. [2021] descriptors, while bond features are encoded using 3-dimensional OGB vectors, augmented with scalar inter-atomic distances for geometric awareness. In addition, a 21-dimensional sequence feature vector (amino acid composition and log-transformed length) and standardized molecular weight (kDa) are included as global descriptors. Both graphs are processed using MPNNs with three layers of NNConv-based message passing and GRU-based updates after each iteration. Each graph is pooled into a fixed-dimensional embedding, while sequence features are encoded using a two-layer MLP. The resulting three embeddings (protein, chromophore, sequence) are concatenated with kDa to form a (3H + 1)-dimensional vector (H = 64), which is passed through a two-layer MLP to predict brightness and $\lambda_{\mathrm{em,max}}$.

All input features are standardized using statistics computed only from the training set, with independent normalization per feature dimension. Target variables are jointly standardized, and predictions are mapped back to physical units for evaluation. Node and edge features are not standardized. Given the limited dataset size (N = 118), a 5-fold cross-validation scheme was adopted to reduce variance in performance estimates. The dataset is shuffled with a fixed seed and partitioned into five disjoint folds of approximately 23–24 proteins each. In each run, one fold is held out as the test set, while the remaining 94–95 proteins are used for training and validation. From this training pool, 17 proteins are selected as a validation set using a fold-specific deterministic shuffle, and the remaining 77–78 proteins are used for training. This procedure ensures that each protein appears exactly once in the test set and multiple times in training across folds, while preserving strict test independence. Final results are reported as mean ± standard deviation across the five folds.

Training minimizes mean squared error in the standardized output space with equal weighting for both targets. Each fold is trained with batch size 8 and optimized using early stopping based on validation loss. Final results are obtained via out-of-fold inference, where each protein is predicted exactly once by a model that did not include it in training.

## 5 Results and Discussion

### 5.1 2D representations

Results of the optimized models are shown in Table 1. For $\lambda_{\mathrm{em,max}}$, ESM outperforms the T-scale + BERT approach, while the T-scale method performs better for brightness prediction. Figure 3 shows predicted versus measured emission for the test data set. Furthermore, *relative* RMSE is large for brightness (brightness $\geq 0.12$); for $\lambda_{\mathrm{em,max}}$, relative RMSE is smaller (range: 400 to 800 nm). Architecture optimization revealed that for ESM, a 2-layer 256-neuron model performs best, while for T-scale a 3-layer 32-neuron model performs best. This stronger performance for deeper architecture suggests that additional non-linear transformations are needed to capture the relationships relevant

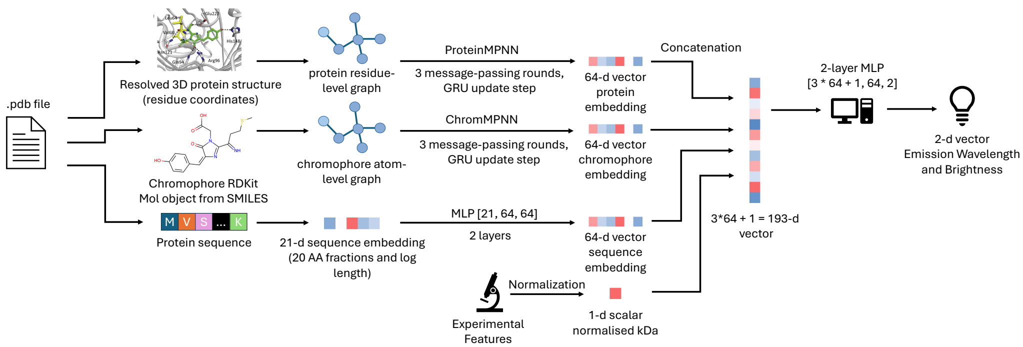


Figure 2: Overview of the machine learning pipeline for the 3D-based model.

for FPs. Training dynamics showed that the ESM model converges in approximately 30 epochs, while T-scale continues improving until the 50th epoch.

Table 1: Prediction error for $\lambda_{\mathrm{em,max}}$ and brightness across sequence embedding strategies.

| | **ESM-2** | | **T-scales + BERT** | |
|---|---|---|---|---|
| | RMSE | MAE | RMSE | MAE |
| Emission wavelength (nm) | 30.46 | 22.65 | 33.82 | 26.23 |
| Brightness | 23.42 | 17.08 | 26.31 | 17.85 |

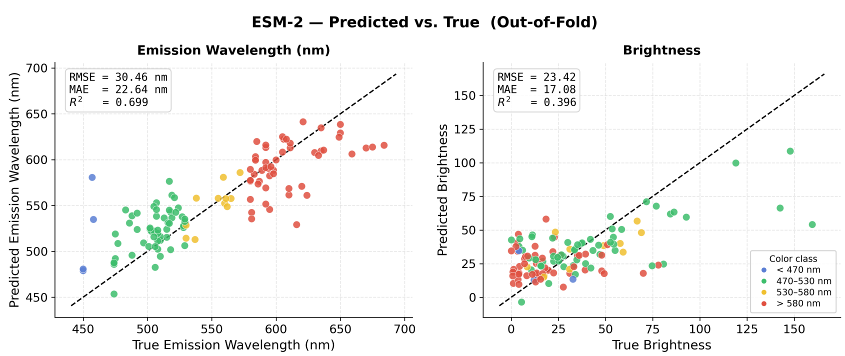


Figure 3: Predicted versus true $\lambda_{\mathrm{em,max}}$ for the ESM-based model.

### 5.2 3D representations

Results per fold for the test sets are given in the docs/documentation.md of the 3D repo. Overall, the MAE for brightness and $\lambda_{\mathrm{em,max}}$ is consistent across folds. Early stopping was applied during training based on validation loss, stopping training once no improvement was observed for a predefined number of epochs. The optimal number of epochs therefore varied across folds. General results for the MPNN-based model are given in Table 2 and Figure 4 shows the predicted vs true $\lambda_{\mathrm{em,max}}$ and brightness for each protein. Similarly to the 2D representations, the qualitative $\lambda_{\mathrm{em,max}}$ (color) is predicted relatively well with an MAE of 28.42 across the 5-fold, for a range of 200 to 800 nm. The pred-vs-true scatter shows a y=x trend, although the model underpredicts at the upper wavelength range. A $\sim$ 28 error for the naive MAE baseline model shows the model is genuinely learning the spread. However, brightness performs very badly. The scatter shows predictions clamped near the predict-mean baseline, so the model does not perform better than a model predicting the mean brightness. The brightness scales between 0 and 100, so an MAE of 21.98 across the folds is still a significant difference that would not make the model reliable. Overall, the model shows better performance on green and red proteins compared to yellow and blue ones, which is likely due to the class imbalance in the dataset, where green and red proteins are more frequently represented during training.

Table 2: Prediction error for $\lambda_{\mathrm{em,max}}$ and brightness for the graph-based model.

| | Predict-mean baseline | 5-fold MAE | Reduction | RMSE |
|---|---|---|---|---|
| Emission wavelength (nm) | 40.90 | $28.42 \pm 2.72$ | $-43\%$ | 35.26 |
| Brightness | 22.51 | $21.98 \pm 4.13$ | $\sim -2\%$ | 30.35 |

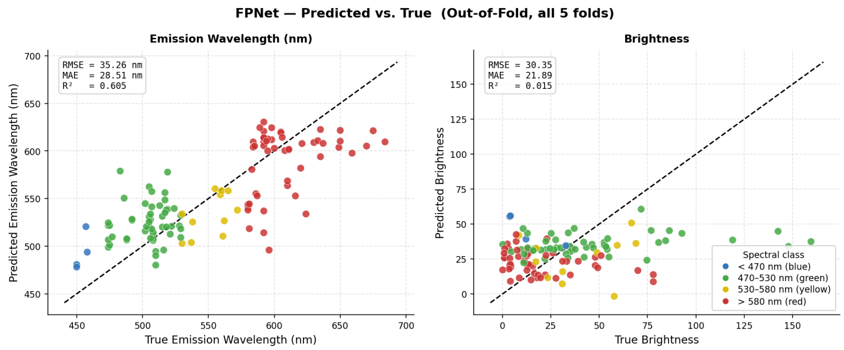


Figure 4: Predicted versus true $\lambda_{\mathrm{em,max}}$ and brightness for the MPNN-based model.

### 5.3 Discussion

For $\lambda_{\mathrm{em,max}}$ and brightness prediction, the 2D model slightly outperforms the 3D model, likely due to its more expressive residue-level feature representation, whereas the 3D model relies primarily on simpler one-hot encoded node features. Both approaches perform substantially better on $\lambda_{\mathrm{em,max}}$ than on brightness. This is expected, as $\lambda_{\mathrm{em,max}}$ is largely determined by the chromophore itself, with surrounding residues providing only fine-tuning effects; thus, explicitly including chromophore information already provides a strong predictive signal. In contrast, brightness is a more complex property that depends on both local and long-range interactions as well as conformational heterogeneity of the protein.

The current models do not explicitly capture coupled chromophore–protein interactions, as the two components are encoded separately rather than jointly within a unified graph. Moreover, the 3D model relies on a single static crystal structure per protein, which does not reflect the dynamic properties of proteins, described as an ensemble of conformations.

Several extensions could improve performance. These include incorporating residue-level features such as ESM-2 embeddings in the 3D model, jointly modeling protein and chromophore within a single graph to enable direct interaction-aware message passing, and accounting for protein dynamics by representing structures as ensembles rather than single conformation (for each protein and each chromophore). Additional improvements could come from explicitly modeling solvent effects in the chromophore pocket (water molecules) and from rebalancing the loss to put more weight on brightness prediction. Most importantly, integrating physics-informed descriptors from quantum chemistry (e.g., DFT-derived orbital energies, HOMO–LUMO gaps, and oscillator strengths) to describe the electronic structure of the dataset, would highly improve both 2D and 3D models.

## 6 Conclusions

Overall, the 2D model slightly outperforms the 3D model, and both perform significantly better on $\lambda_{\mathrm{em,max}}$ than on brightness. This reflects that emission is more directly determined by the chromophore, while brightness depends on more complex structural and environmental effects that are not fully captured by either representation. Future improvements would mainly benefit from better modeling of protein–chromophore interactions and, most importantly, a larger dataset of FPs to improve generalization.

## 7 Comment on the use of AI

The python code utilized for this project was crafted with the assistance of several LLMs. They were used for clarification or understanding (debugging, assistance with mathematical misconceptions about algorithms), or in a generative way (adaptation of code found in Sources like StackOverflow or taken from this course's exercise sessions; *de novo* generation of larger portions of code tailored to our aims). Prompt examples are given hereafter:

1. Debugging: "Explain why this crashes [*quotes code that crashes*]"
2. Mathematical assistance: "My current input is of shape nxn where shape mxm is required, is there a way to transform my input to mxm shape?"
3. Adaptation: "How do I implement this scaler tool [*quotes scaler tool found on StackOverflow*] in my current encoding tool [*quotes python code for encoding tool*]?"
4. Generation: "I would like to use 3d crystallographic data from PDB as input into a neural network. Come up with an python code for an encoder that fetches data on PDB and transforms it into a format that can be fed into a neural network."

For 1-3, OpenAi's ChatGPT was used. For 4, Anthropic's Claude was used. While for Modes 1-3, the overall pipeline of the model remains under control, mode 4 requires reflection on whether the result actually matches the purpose and general idea of the project.

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